EVERY PRIME DIVIDES A TERM OF OEIS A363414

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ABSTRACT. Let $a(n) = \frac{1}{2}\operatorname{Im}\prod_{k=0}^{n}(1+2ki)$ (OEIS A363414). Bala conjectured that every prime divides some term $a(n)$ with $n > 0$, that is, that there are no primes of "type 1" for this sequence. We prove this: $p \mid a(p-1)$ for every odd prime p, and $a(3) = -18$. The proof is two lines. It is formalized in Lean 4 with Mathlib, against the statement `OeisA363414.conjecture_type1_empty` in Google DeepMind's Formal Conjectures repository.

1. THE RESULT

Let $P_n = \prod_{k=0}^{n}(1+2ki) \in \mathbb{Z}[i]$. Every factor is $\equiv 1 \pmod{2\mathbb{Z}[i]}$, so $P_n \equiv 1 \pmod 2$ and $\operatorname{Im} P_n$ is even. The sequence A363414 [2] is $a(n) = \frac{1}{2}\operatorname{Im} P_n$; it begins $0, 1, 3, -18, -190, \ldots$. Following Moll's classification of primes for related sequences, Bala calls a prime of *type 1* if it divides no term $a(n)$ with $n > 0$. He conjectured that there are no type 1 primes. The conjecture is stated in [1] as `OeisA363414.conjecture_type1_empty`: for every prime p there is $n > 0$ with $p \mid a(n)$.

Theorem 1. *For every odd prime p, $p \mid a(p-1)$. Moreover $a(3) = -18$. Hence every prime divides some $a(n)$ with $n > 0$.*

Proof. Let p be odd and work in $R = \mathbb{Z}[i]/p\mathbb{Z}[i]$. Reversing the order of the factors, $k \mapsto p-1-k$, gives

$$P_{p-1} = \prod_{k=0}^{p-1}\big(1+2(p-1-k)i\big) \equiv \prod_{k=0}^{p-1}\big(1-2(k+1)i\big) = \prod_{j=1}^{p}(1-2ji) \equiv \prod_{j=0}^{p-1}(1-2ji) = \overline{P_{p-1}} \pmod p.$$

The second congruence holds because the factor for $j = p$ is $\equiv 1$, as is the factor for $j = 0$. Hence p divides $P_{p-1} - \overline{P_{p-1}} = 2i\operatorname{Im} P_{p-1} = 4i\,a(p-1)$ in $\mathbb{Z}[i]$. Since p is odd, $p \mid a(p-1)$, and $p - 1 \geq 2 > 0$. Finally, $P_3 = (1+2i)(1+4i)(1+6i) = -43-36i$, so $a(3) = -18$, which is even. □

Remark 2. The value of P_{p-1} modulo p can be computed exactly. In $\mathbb{F}_p[X]$ we have $\prod_{t\in\mathbb{F}_p}(X - t) = X^p - X$. Substitute $X = -\iota^{-1}$, where ι is the image of i in R, and multiply by $(-\iota)^p$. This gives $\prod_{t\in\mathbb{F}_p}(1 + t\iota) = 1 - \iota^{p-1} = 1 - (-1)^{(p-1)/2}$. As $2k$ runs over all residues modulo p, we get $P_{p-1} \equiv 1 - (-1)^{(p-1)/2} \pmod p$, which is 0 for $p \equiv 1$ and 2 for $p \equiv 3 \pmod 4$. The same computation with $\sqrt{-d}$ in place of i gives the analogous statement for $\prod(1 + k\sqrt{-d})$ and every prime $p \nmid 2d$.

Remark 3. The companion conjecture `conjecture_type2_asymptotics`, that $v_p(a(n)) \sim n/(p-1)$ for $p \equiv 1 \pmod 4$, is not addressed here.

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2. Formalization

The theorem is formalized in Lean 4 with Mathlib. The statement file of [1] is included unmodified. `OeisA363414.conjecture_type1_empty_solved` has the type of the repository statement, which is checked by `rfl`. The formal proof follows the reversal argument above. It maps the complex product to the Gaussian integers, and it evaluates $a(3)$ with `norm_num`. It depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. Independently we checked the congruence $P_{p-1} \equiv 1 - (-1)^{(p-1)/2} \pmod{p}$ for all odd primes $p < 20000$, and $p \mid a(p-1)$ with exact Gaussian integers for $p < 400$.

References

[1] Google DeepMind, *Formal Conjectures*, https://github.com/google-deepmind/formal-conjectures; the statement is in `FormalConjectures/OEIS/363414.lean`, pull request 4946 (2026).

[2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org/A363414.